

Software Engineering Department

Braude College of Engineering

Final Project - Phase A (61998)

AI Agent for Clinic Operational Management

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GitHub Link:
https://github.com/AmirAzulay/AI-Agent-for-Clinic-Operational-Management

Abstract

Medical clinics operate in highly dynamic environments where patient arrivals, resource availability, and operational workloads continuously change throughout the day. These fluctuations can create operational bottlenecks, increase patient waiting times, and reduce the efficiency of resource utilization. Traditional clinic management systems primarily provide data displays and reports, but they lack the ability to understand operational context, identify emerging issues in real time, or proactively recommend corrective actions. As a result, clinic staff are required to manually monitor numerous metrics and dashboards, detect anomalies on their own, and make decisions under pressure and uncertainty. This approach can lead to delayed responses to problems, suboptimal resource allocation, and a strong dependence on the experience and availability of personnel.

This project proposes an AI-powered operational management system for a simulated medical clinic environment. The system combines a real-time clinic simulation with an AI Agent capable of analyzing operational conditions, identifying bottlenecks, detecting workload imbalances, and autonomously executing predefined management actions through a controlled Tool Calling architecture. The main idea of the proposed approach is the use of an autonomous AI agent that continuously observes the clinic state, reasons about operational conditions, and decides when and how to act. Unlike conventional rule-based dashboards or monitoring systems, the agent can combine information from multiple sources, understand the overall situation in the clinic, prioritize issues, and dynamically select appropriate actions. Instead of just displaying information, it functions as an intelligent management layer that transforms raw operational data into actionable decisions.

The use of an AI agent is particularly important because clinic operations involve many interconnected variables that change simultaneously. Human operators may overlook hidden patterns or emerging risks when monitoring multiple dashboards, especially during busy periods. An AI agent can continuously analyze the environment, detect trends before they become critical problems, and respond consistently according to predefined policies. This enables faster reaction times, reduces the pressure on staff, and supports more efficient and scalable operational management.

The proposed solution focuses on supporting operational decision-making and does not involve medical diagnosis. By monitoring patient queues, treatment room availability, physician workloads, and medication inventory levels, the AI agent can identify operational issues, generate alerts, initiate predefined corrective actions, and provide clear, explainable operational summaries for clinic staff and managers. The objective of this project is to demonstrate how AI Agent technologies can support the management and operation of healthcare systems through intelligent monitoring, controlled decision-making, and automated operational assistance. The expected outcome is improved clinic operational efficiency, reduced reliance on manual monitoring, optimized resource utilization, and enhanced ability to handle operational challenges within a simulated clinic environment. Furthermore, the project demonstrates how artificial intelligence can be integrated as an active operational mechanism rather than just a reporting tool, thereby reducing managerial workload, improving response times to events, and enabling staff to focus on higher-value tasks.

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1 - Introduction

Healthcare clinics operate in dynamic environments where patient arrivals, staff availability, treatment room occupancy, and operational resources continuously change throughout the day. Efficient management of these factors is essential for reducing waiting times, improving resource utilization, and maintaining service quality [5]. Traditional clinic operations often rely on human personnel to monitor patient queues, identify overloaded departments, allocate resources, and respond to operational issues. As clinic complexity increases, this approach can lead to delayed responses, information overload, and inefficient decision-making [3],[5].

Healthcare organizations commonly use Healthcare Information Systems (HIS), Electronic Health Record (EHR) platforms, and Clinic Management Systems (CMS) to support administrative and clinical workflows. Systems such as Epic, Oracle Health, and athenaOne provide capabilities including patient registration, appointment scheduling, medical record management, resource tracking, reporting, and operational coordination. While these platforms improve data accessibility and workflow management, they primarily focus on collecting, organizing, and presenting information. The responsibility for interpreting operational conditions and determining appropriate responses generally remains with human operators [5].

Recent advances in Artificial Intelligence have expanded the use of intelligent decision-support systems within healthcare environments. AI technologies have demonstrated potential in operational areas such as patient flow optimization, scheduling support, workforce management, administrative automation, and resource allocation [1]. Unlike traditional software systems, AI Agents can analyze complex situations, reason about operational conditions, and select actions through interactions with predefined tools [4],[8] Research on ReAct and Toolformer demonstrates how language models can combine reasoning with action execution and make effective use of external tools to solve operational tasks [4],[8]

Simulation is widely used for evaluating healthcare processes because it enables the study of dynamic environments without relying on real patient data. Discrete-event simulation is commonly applied to represent patient arrivals, queue formation, treatment durations, room occupancy, and resource utilization in healthcare settings [2],[6]. Simulation environments provide a safe and repeatable way to create operational scenarios such as department overload, long waiting times, medication shortages, and resource constraints, making them suitable for evaluating AI-driven operational management systems [2],[6].

The proposed project combines these concepts by developing an AI-powered operational management system for a simulated clinic environment. The system utilizes an AI Agent that continuously monitors operational conditions, identifies bottlenecks, and performs predefined management actions through a controlled tool-calling architecture. The agent is restricted to operational decision support and does not perform medical diagnosis or treatment recommendations, following recommendations for safe and explainable healthcare AI systems [7]. By integrating simulation, operational monitoring, AI reasoning, and controlled tool execution, the project aims to bridge the gap between traditional clinic management systems that primarily display information and intelligent systems capable of providing proactive operational assistance [4],[8].

2 - Background and Related Work

2.1 Healthcare Operational Management and Patient Flow

Healthcare clinics operate in dynamic environments where patient arrivals, waiting queues, treatment rooms, medical staff, and operational resources continuously change throughout the day. Efficient operational management is essential for maintaining acceptable waiting times, reducing congestion, improving patient satisfaction, and maximizing resource utilization. Patient flow is influenced by multiple factors, including arrival rates, treatment duration, room availability, physician workload, department capacity, urgency levels, and administrative processes [3],[5].

Traditionally, clinic operations rely heavily on human personnel to monitor queues, track resource availability, identify bottlenecks, and respond to operational issues. Reception staff, nurses, and managers are often required to monitor multiple data sources simultaneously while making decisions under time pressure. Although this approach may be effective in smaller clinics, it becomes increasingly difficult as patient volume and operational complexity grow. Delayed responses, information overload, and inefficient resource allocation may negatively impact clinic performance and patient satisfaction [3],[5].

Recent research shows that artificial intelligence is increasingly being applied not only to clinical diagnosis but also to healthcare operations. AI technologies are being used to improve scheduling, patient flow management, workforce planning, and resource allocation. These developments demonstrate a growing interest in intelligent operational support systems within healthcare environments.

2.2 Existing Healthcare Management Systems

Healthcare organizations commonly use Healthcare Information Systems (HIS), Electronic Health Record (EHR) platforms, and Clinic Management Systems (CMS) to support daily clinical and administrative workflows. Examples of such systems include Epic, Oracle Health, and athenaOne.

Several commercial platforms are widely adopted throughout the healthcare industry. Epic Systems provides a comprehensive healthcare platform centered around patient records and operational workflows. Oracle Health offers tools for patient flow management, resource utilization, length-of-stay monitoring, discharge coordination, and operational visibility. Similarly, athenaOne provides an integrated cloud-based platform that combines EHR functionality, medical billing, patient engagement, and practice management capabilities

Although these systems significantly improve information accessibility and workflow coordination, they primarily focus on collecting, organizing, and presenting operational data. In most cases, healthcare personnel remain responsible for interpreting information, identifying operational problems, and deciding which actions should be taken. Consequently, there remains a gap between information management systems and intelligent systems capable of actively supporting operational decision-making.

The proposed project addresses this gap by introducing an AI Agent that continuously monitors clinic conditions, analyzes operational situations, and performs predefined actions

through a controlled tool-calling architecture. Rather than functioning solely as a reporting system, the proposed solution provides proactive operational support.

2.3 Artificial Intelligence in Healthcare Operations

Artificial Intelligence has become an important research area in healthcare. While many AI applications focus on medical diagnosis, disease prediction, medical imaging, and clinical risk assessment, AI is increasingly being used for operational and administrative tasks [1].

AI-based operational systems can process large volumes of data and identify patterns that may be difficult for human operators to detect in real time. For example, AI can detect increasing waiting times, identify overloaded departments, recognize resource shortages, and support scheduling decisions. Such capabilities allow healthcare organizations to improve responsiveness, reduce manual workload, and optimize resource utilization [4],[5].

Nasef et al discuss the integration of AI into hospital management and healthcare operations, highlighting opportunities for workforce planning, capacity management, and operational optimization. Similarly, Varnosfaderani et al [5] describes the growing role of AI in improving hospital efficiency, administrative workflows, patient flow, and scheduling processes.

In the context of this project, AI is not used to replace healthcare professionals or perform medical diagnosis. Instead, the AI Agent is restricted to operational decision support. The agent analyzes clinic conditions such as patient queues, waiting times, room availability, physician workload, department status, and medication inventory. Based on this information, it can identify operational bottlenecks, generate alerts, and initiate predefined actions designed to improve clinic efficiency.

2.4 Simulation-Based Healthcare Management

Simulation is widely used for analyzing and improving healthcare systems because healthcare environments are complex, dynamic, and difficult to evaluate directly in real-world settings. Discrete-event simulation (DES) is one of the most common approaches used for modeling healthcare operations because it can represent patient arrivals, waiting queues, treatment durations, room occupancy, and resource utilization [2],[6].

Forbus & Berleant [2] reviewed the use of discrete-event simulation in healthcare and demonstrated its application in healthcare operations, capacity planning, patient flow management, and resource allocation. Similarly, Vázquez-Serrano et al [6] describe DES as an effective approach for evaluating complex healthcare processes and testing operational improvements before implementation.

Simulation is particularly suitable for this project because it enables realistic testing without relying on real patient data. A simulated clinic environment can generate controlled scenarios such as department overload, long waiting times, medication shortages, physician overload, and resource constraints. These scenarios provide a safe environment for evaluating AI-driven operational decision-making while avoiding privacy, safety, and ethical concerns associated with real healthcare systems.

An additional advantage of simulation is repeatability. The same operational scenario can be executed multiple times, allowing the behavior of the AI Agent to be tested, evaluated, and improved under controlled conditions.

2.5 AI Agents, Tool Calling, and System Safety

AI Agents represent an evolution beyond traditional chatbots and monitoring systems. While conventional systems primarily provide information or textual responses, AI Agents can reason about situations, determine appropriate actions, and interact with external tools to achieve operational goals.

The ReAct framework introduced by Yao et al [8] combines reasoning and acting within language models. Instead of generating a single response, the model can analyze a situation, decide on an action, execute that action, and continue reasoning based on the results. This approach improves transparency, adaptability, and decision-making capabilities.

Tool former extends this concept by demonstrating how language models can learn to use external tools and APIs when necessary [4]. Rather than relying only on internal knowledge, the model can access external functions, retrieve information, and perform structured actions through predefined interfaces.

These concepts are directly relevant to the proposed project. The clinic management system adopts controlled tool-calling architecture in which the AI Agent continuously analyzes clinic conditions and selects actions using predefined backend tools. Examples include reading clinic status, opening inactive treatment rooms, creating alerts, updating department status, requesting medication restocking, and generating operational summaries.

A key design principle is the separation between reasoning and execution. The AI Agent determines which action should be performed, while the backend controls how the action is executed. This architecture improves safety, transparency, maintainability, and system reliability because all actions can be validated, monitored, and logged before execution [4],[8].

Healthcare-related AI systems must also address explainability, safety, and human oversight. The World Health Organization emphasizes that AI systems in healthcare should be transparent, accountable, safe, and designed to support human decision-makers rather than replace them [7].

For this reason, the proposed system limits the AI Agent to predefined operational actions only. The agent cannot diagnose diseases, recommend treatments, prescribe medications, or modify clinical records. All actions are logged and accompanied by explanations that describe why the action was performed. This approach supports transparency, enables human review, and helps build trust in the system while maintaining safe operational boundaries [7].

3 - Project Requirements

FR	NFR
The system shall simulate patient arrivals at predefined time intervals during the clinic working day.	The dashboard shall provide a clear and intuitive user interface.
The system shall create and maintain patient records during the simulation.	The system shall display clinic information in an organized and readable manner.
The system shall store patient information including department assignment, urgency level, waiting time, and treatment status.	The system shall update operational information with minimal delay during the simulation.
The system shall assign each patient to a relevant clinic department according to predefined simulation rules.	System operations and AI actions shall be consistently recorded.
The system shall support multiple clinic departments such as general medicine, pediatrics, orthopedics, and emergency care.	The AI Agent shall provide explanations for all executed actions.
The system shall manage clinic departments and their operational states.	AI Agent explanations shall be understandable to non-technical users.
The system shall maintain the operational status of each department, such as Normal, Warning, Overloaded, and Critical.	Access to the system shall be restricted to authorized users.
The system shall track the number of patients waiting in each department queue.	The system shall support user authentication for accessing the dashboard.
The system shall calculate and update patient waiting time throughout the simulation.	The system shall protect operational data from unauthorized access.
The system shall manage treatment rooms and room availability.	The architecture shall support future expansion and modification.
The system shall track whether each treatment room is available, occupied, inactive, or under maintenance.	Additional departments, rooms, doctors, medications, and users shall be supported without major redesign.
The system shall allow inactive treatment rooms to be opened when overload conditions are detected.	The system shall allow simulation parameters to be modified without changing the core architecture.
The system shall track doctor availability for each department.	The system shall be modular and separate the dashboard, backend, database, simulation engine, and AI Agent components.
The system should track doctor's workload according to assigned patients and active treatments.	The backend shall expose structured APIs for communication between system components.

The system shall manage medication inventory levels.	The system shall store data in a structured and consistent database format.
The system shall detect low medication inventory when stock falls below a predefined threshold.	The system shall maintain reliable operation during continuous clinic simulation.
The system shall generate medication restock requests when medication inventory is low.	The system should provide consistent response formats for backend API responses.
The system shall generate operational alerts when abnormal situations occur.	The AI Agent shall be restricted to predefined operational tools only.
The system should generate alerts for long waiting times, department overload, room shortage, doctor overload, and medication shortage.	The AI Agent shall not perform medical diagnosis or treatment recommendations.
The system should provide a dashboard for clinic monitoring.	The system shall log errors and important events for debugging and evaluation.
The dashboard shall display patient queues, department statuses, room availability, doctor workload, medication inventory, and active alerts.	The system shall support integration testing between the dashboard, backend, database, simulation engine, and AI Agent.
The system allows authorized users to activate the AI Agent.	The system should be maintainable and understandable for future development.
The AI Agent shall read the current clinic state before making operational decisions.	The system shall provide stable performance under predefined simulated clinic workloads.
The AI Agent shall analyze clinic conditions and identify operational bottlenecks.	The system shall be containerization-ready to support future deployment using Docker.
The AI Agent shall select predefined operational tools according to the detected clinic condition.	The system should be designed to allow future integration with external clinic management systems.
The AI Agent shall be able to create operational alerts through a controlled backend tool.	
The AI Agent shall be able to update department status through a controlled backend tool.	
The AI Agent shall be able to open an inactive treatment room through a controlled backend tool.	
The AI Agent shall be able to generate medication restock requests through a controlled backend tool.	
The AI Agent shall generate an operational summary describing the current clinic state, detected problems, and performed actions.	

4 - Research and Development Process

4.1 Research Phase

The research phase will focus on studying:

- AI Agent architectures
- Tool-calling frameworks
- Decision-support systems
- Healthcare operational management systems
- Simulation-based management environments

Relevant academic papers, technical documentation, and industry solutions will be reviewed in order to define the system architecture and project scope.

4.2 Development Methodology

The project will be developed incrementally using an Agile development approach. Development will proceed through the following stages:

1. Requirements analysis
2. Database design
3. Backend implementation
4. Clinic simulation implementation
5. Dashboard development
6. AI Agent implementation
7. Integration testing
8. System evaluation

4.3 AI Agent Architecture and Design

The AI Agent is the core component of the proposed system and is responsible for supporting operational decision-making within the simulated clinic environment. Unlike a traditional chatbot, which generates textual responses to user queries, the AI Agent continuously monitors the clinic, analyzes operational conditions, determines whether intervention is required, and performs predefined operational actions through a controlled tool-calling architecture.

The agent is designed as an autonomous decision-support component rather than a standalone language model. Its architecture separates reasoning from execution by combining structured operational data, prompt engineering, Large Language Model (LLM) reasoning, function calling, backend validation, and system logging. This design ensures that every decision is explainable, controlled, and aligned with the operational rules of the clinic.

4.3.1 Context Builder

Every decision cycle begins with collecting the current operational state of the clinic.

A dedicated **Context Builder** retrieves information from the backend and database, including:

- Patient queues for each department
- Average waiting times
- Department operational status
- Treatment room availability
- Doctor workload
- Medication inventory
- Active operational alerts
- Previous AI actions

Instead of sending the complete database to the AI model, the Context Builder creates a structured JSON object containing only the information required for decision-making. This approach reduces unnecessary data, improves efficiency, and ensures that the AI Agent receives a consistent representation of the clinic state during every reasoning cycle.

4.3.2 Prompt Engineering

The Prompt Engineering layer defines the behavior, responsibilities, and operational boundaries of the AI Agent.

A predefined system prompt instructs the model to:

- Analyze the clinic's operational state.
- Detect operational bottlenecks.
- Select only predefined backend tools.
- Never perform medical diagnosis or treatment recommendations.
- Return structured JSON responses.
- Generate a short explanation for every selected action.

4.3.3 Decision-Making Process

The reasoning engine is implemented using a **Large Language Model (LLM)** accessed through an API, such as **OpenAI**, **Claude**, or **Gemini**.

After receiving the system prompt and the current clinic context, the model analyzes the operational conditions and determines whether intervention is required.

Unlike rule-based systems that rely on predefined if-else conditions, the LLM evaluates the complete operational context before deciding which action is most appropriate.

4.3.4 Tool Calling and Backend Validation

The AI Agent never interacts directly with the database or modifies the clinic simulation.

Instead, it communicates with the system through predefined backend tools using a **Function Calling** mechanism.

The LLM returns a structured JSON object containing:

- The selected tool
- Required parameters
- A short explanation describing the reason for the decision

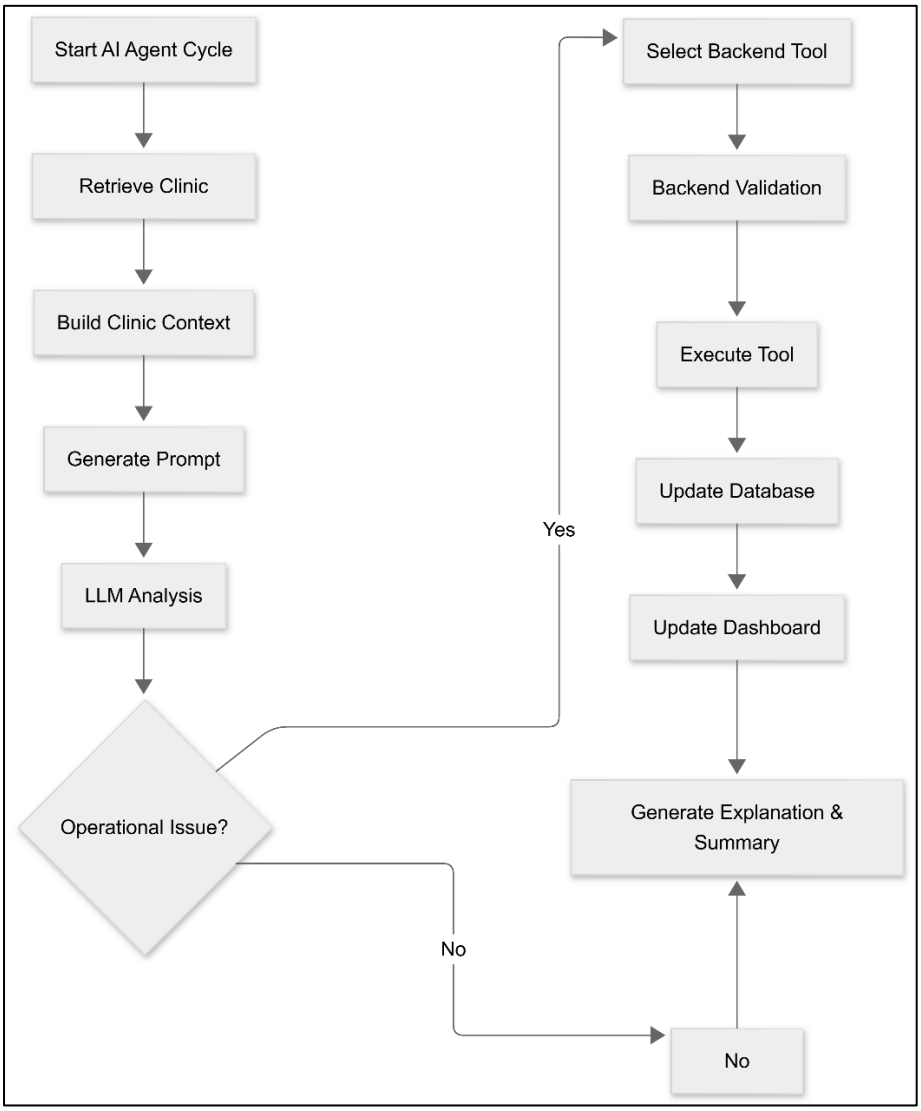
Before executing any operation, the backend validates:

- The selected tool
- Input parameters
- Current clinic state
- Business rules and operational constraints

Only validated requests are executed. This architecture guarantees that the AI Agent remains predictable, secure, and fully controlled by the backend.

4.3.5 AI Agent Workflow

1. The backend retrieves the latest clinic data from the simulation and database.
2. The Context Builder creates a structured JSON representation of the current clinic state.
3. The system combines the clinic context with the predefined system prompt.
4. The prompt is sent to the selected LLM for reasoning.
5. The model analyzes the clinic state and determines which tool to execute.
6. The backend validates the selected tool and its parameters.
7. The validated action is executed and the database is updated.
8. The dashboard is refreshed with the latest operational information.
9. The AI Agent generates an explanation and an operational summary describing the detected problem, the selected action, and the current clinic state.



4.3.6 AI Tools and Tool Selection

The AI Agent performs actions exclusively through predefined backend tools. Restricting the agent to a fixed set of tools improves security, transparency, and predictability while preventing unauthorized operations.

During every reasoning cycle, the AI Agent receives the current clinic context together with the list of available tools. Based on the operational data and the predefined system prompt, the LLM determines whether intervention is required. If a problem is detected, it selects the single tool that best addresses the current situation and returns the required parameters using the Function Calling mechanism.

The backend validates every request before execution, ensuring that only legal and appropriate actions are performed.

Tool	Purpose	When the Agent Uses It	Expected Output
READ_CLINIC_STATE	Retrieves the current clinic state.	At the start of each cycle.	Clinic data (JSON).
CREATE_ALERT	Creates an operational alert.	When an issue requires attention.	New alert.
SET_DEPARTMENT_STATUS	Updates the status of a department.	When department conditions change.	Updated department status.
OPEN_ROOM	Opens an inactive treatment room.	When additional room capacity is needed.	Active treatment room.
REQUEST_MEDICATION_RESTOCK	Creates a medication restock request.	When medication stock is low.	Restock request.
GENERATE_SUMMARY	Creates an operational summary.	After all actions are completed.	Summary report.

4.4 Expected Outputs

The proposed system is expected to generate several outputs that support operational decision-making and clinic management:

- **Operational Alerts** - Real-time notifications for situations such as department overload, long waiting times, room shortages, and low medication inventory.
- **Department Status Updates** - Dynamic classification of departments based on their current operational condition (e.g., Normal, Warning, Overloaded, Critical).
- **AI Action Logs** - Detailed records of all actions performed by the AI Agent, including the reason for each decision and its outcome.
- **Medication Restock Requests** - Automatic requests generated when medication inventory falls below predefined thresholds.
- **Operational Summaries** - AI-generated summaries describing clinic conditions, detected issues, and actions performed.

5 - Expected Challenges

Several challenges are anticipated during the development of the proposed system. Since the project combines a clinic simulation, backend services, a database, a dashboard, and an AI Agent, the main challenges involve creating a realistic simulation, ensuring reliable AI decision-making, synchronizing system components, evaluating system performance, and maintaining safe and explainable AI behavior.

5.1 Designing a Realistic Clinic Simulation

A major challenge is developing a simulation that realistically represents clinic operations while remaining manageable within the project's scope. The simulation should generate dynamic scenarios such as patient arrivals, waiting queues, department overloads, room shortages, and medication shortages to provide meaningful conditions for evaluating the AI Agent.

5.2 AI Decision-Making and Tool Selection

The AI Agent must accurately analyze the clinic state and select the most appropriate operational action. Since the agent interacts with backend tools, it must avoid unnecessary or incorrect actions. All decisions will therefore be limited to predefined tools and validated before execution to ensure safe and predictable behavior.

5.3 System Integration and Synchronization

The proposed system consists of multiple components, including the simulation engine, backend, database, dashboard, and AI Agent. Ensuring reliable communication and synchronization between these components is essential so that the AI Agent always makes decisions based on the latest clinic state.

5.4 Evaluating AI Performance

Another challenge is defining objective criteria for evaluating the AI Agent. The system will be assessed according to its ability to detect operational bottlenecks, select appropriate actions, generate meaningful explanations, and improve clinic efficiency under different simulation scenarios.

5.5 Explainability and Project Scope

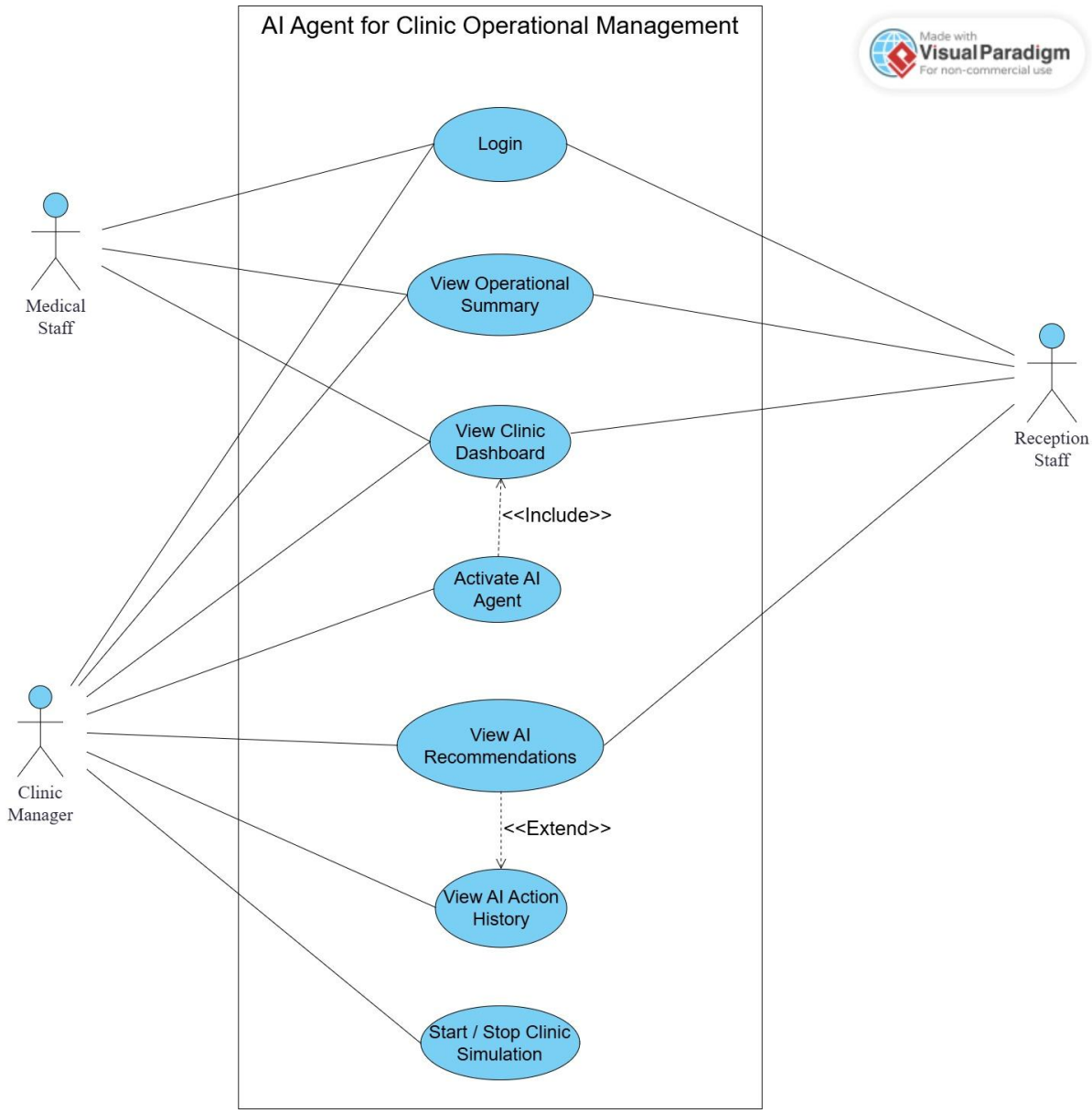
The AI Agent should provide clear explanations for every action to improve transparency and user trust. At the same time, the project must remain focused on demonstrating AI-assisted operational management in a simulated clinic, avoiding unnecessary features that are beyond the scope of the project, such as medical diagnosis or treatment recommendations.

6 - Technologies and Tools

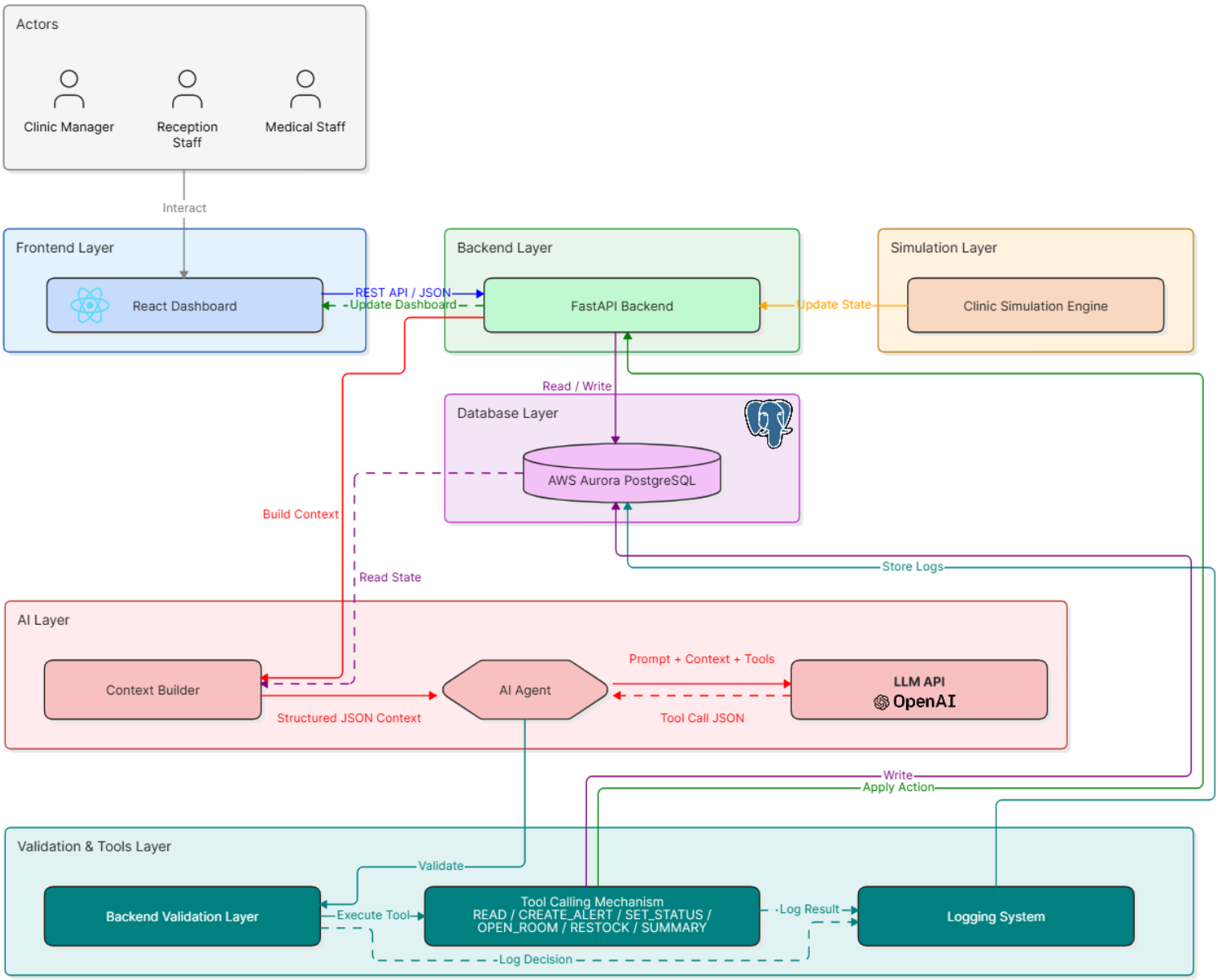
Component	Technology
Frontend	React
Backend	FastAPI (Python)
Database	AWS Aurora PostgreSQL
AI Integration	OpenAI/Claude/Gemini Models
API Communication	REST APIs
Version Control	Git & GitHub
Containerization	Docker

7 - Diagrams (UC, Activity, Architecture)

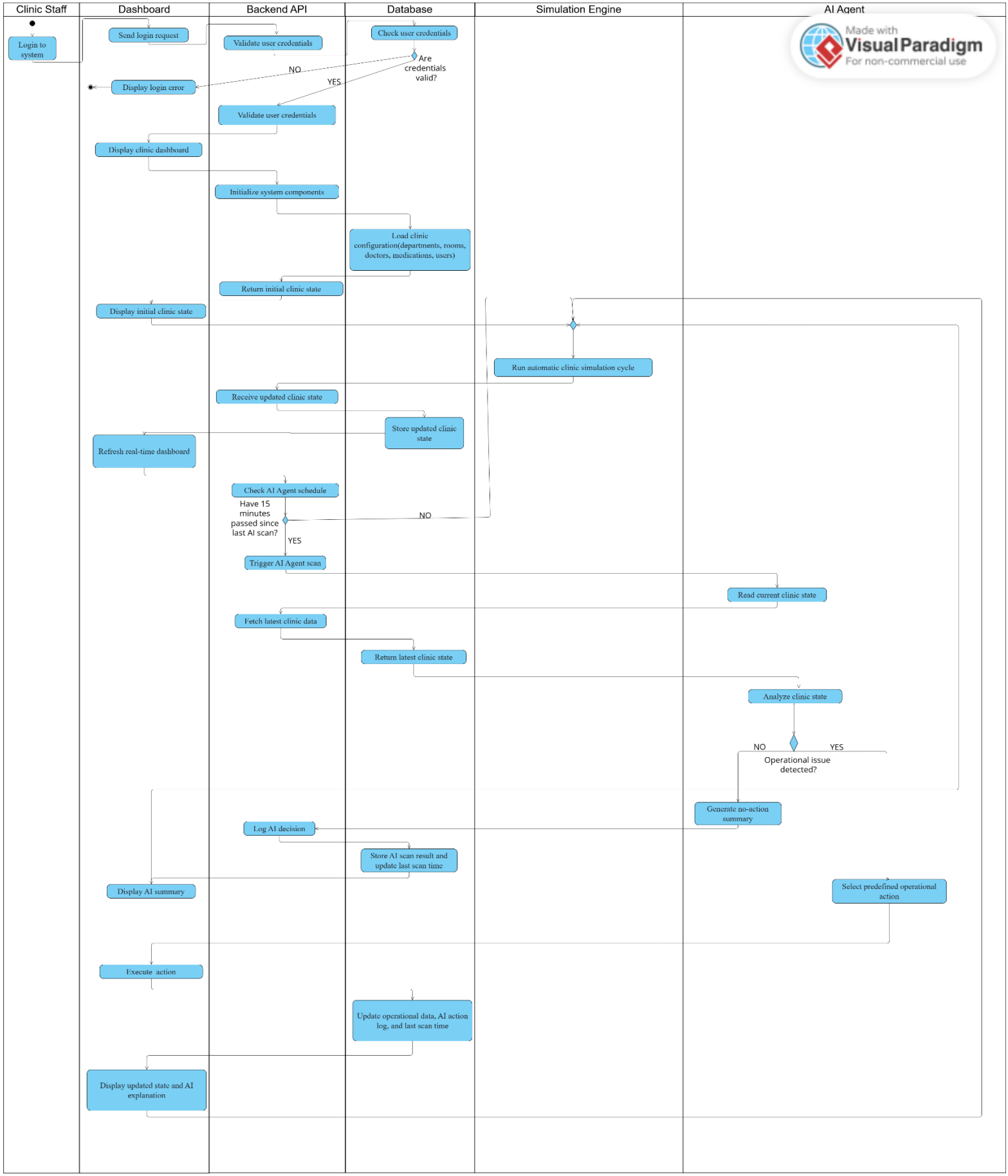
Use Case Diagram:



Architecture Diagram:



Activity Diagram:



8 - Project Success Metrics

The objective of the project is to demonstrate that an AI Agent can effectively support the operational management of a simulated medical clinic by continuously monitoring clinic conditions, identifying operational bottlenecks, and executing predefined corrective actions through a controlled tool-calling architecture.

8.1 Operational Issue Detection Accuracy

This metric measures how well the AI Agent detects operational problems such as department overload, long waiting times, room shortages, and medication shortages.

How it will be measured:
Several simulation scenarios will be created with known operational problems. The number of correctly detected issues will be compared with the total number of expected issues.

Detection Accuracy = (Correctly Detected Issues / Total Expected Issues) × 100
Success Target: At least **90%**. This means the AI Agent should detect almost all operational problems during the simulation.

8.2 Corrective Action Accuracy

This metric measures whether the AI Agent selects the correct action after detecting a problem.

How it will be measured:
For each test scenario, the expected action will be defined in advance. The selected action will then be compared with the expected one.

Action Accuracy = (Correct Actions / Total Actions Executed) × 100
Success Target: At least **85%**. This means the AI Agent should choose the correct action in most situations.

8.3 Incorrect Action Rate

This metric measures how often the AI Agent performs unnecessary or incorrect actions.

How it will be measured:
After each simulation, the executed actions will be compared with the expected behavior. Actions that are unnecessary or inappropriate will be counted as incorrect.

Incorrect Action Rate = (Incorrect Actions / Total Actions Executed) × 100
Success Target: Less than **10%**. This means the AI Agent should make very few unnecessary or incorrect actions.

8.4 Explanation Coverage

This metric measures whether the AI Agent explains every action it performs.

How it will be measured:
The system logs will be checked after each simulation to verify that every AI action includes an explanation.

Explanation Coverage = (Actions with Explanations / Total Actions Executed) × 100
Success Target: 100%. Every action performed by the AI Agent should include an explanation so users can understand why it was taken.

9 - Testing Process

The system will be tested using a simulated clinic environment that generates predefined operational scenarios. The purpose of the testing process is to verify that the AI Agent can correctly identify operational issues, select appropriate actions, execute predefined tools, and provide clear explanations for its decisions.

Test	Expected Result
Log in with valid credentials.	The user is successfully logged in and the dashboard is displayed.
Log in with an invalid username or password.	An error message is displayed and access is denied.
Start the clinic simulation.	Patient generation begins and the dashboard is updated in real time.
Generate patients for a department.	New patients are added to the selected department queue.
Create a department overload.	The department status changes to Overloaded and an alert is generated.
Create a medication shortage.	A medication shortage alert is generated and a restock request is created.
Occupy all treatment rooms in a department.	The system detects a room shortage.
Activate the AI Agent.	The AI Agent reads the current clinic state and starts monitoring the simulation.
Generate a department overload while the AI Agent is active.	The AI Agent detects the overload and selects the appropriate corrective action.
Execute the OPEN_ROOM tool.	An inactive treatment room is activated successfully.
Execute the CREATE_ALERT tool.	A new operational alert appears on the dashboard.
Execute the SET_DEPARTMENT_STATUS tool.	The selected department status is updated correctly.
Execute the REQUEST_MEDICATION_RESTOCK tool.	A medication restock request is created successfully.
Execute the GENERATE_SUMMARY tool.	An operational summary describing the clinic state and AI actions is generated.
Perform an AI action.	The action is recorded in the system log together with its explanation.
Run the simulation without operational problems.	The AI Agent does not perform unnecessary actions or generate false alerts.
Run all system components together.	The dashboard, backend, database, simulation engine, and AI Agent communicate correctly without synchronization errors.

10 - AI Tools and Prompts

AI Tools Used

During the research, planning, system design, and documentation phases of the project, we used **ChatGPT (OpenAI)** as an AI-assisted research and development support tool.

Tool Used:

- ChatGPT (OpenAI) - https://chatgpt.com

Purpose of Use:

- Literature review assistance
- Analysis of existing healthcare management systems
- Requirement engineering support
- System architecture design
- AI Agent workflow design
- Research and development planning
- Identification of project challenges and risks
- Definition of success metrics and testing scenarios
- Academic writing and document refinement

Representative Prompts Used During the Project

Prompt 1 - Literature Review

Provide a literature review on the use of Artificial Intelligence and AI Agents in healthcare operational management, focusing on patient flow optimization, resource allocation, scheduling, and decision-support systems.

Prompt 2 - Existing Solutions Analysis

Analyze existing healthcare management systems such as Epic, Oracle Health, and athenaOne. Identify their strengths, limitations, and gaps regarding autonomous operational decision-making.

Prompt 3 - AI Agent Architecture Design

Design a high-level architecture for an AI-powered clinic operational management system. Include a dashboard, backend, database, simulation engine, AI Agent, and tool-calling mechanism.

Prompt 4 - Functional Requirements

Generate functional requirements for a software engineering project called "AI Agent for Clinic Operational Management". The system should include a clinic simulation, operational monitoring, and an AI Agent capable of executing predefined operational actions.

Prompt 5 - Non-Functional Requirements

Generate non-functional requirements for an AI-powered clinic management system, including performance, scalability, security, maintainability, reliability, and explainability requirements.

Prompt 6 - AI Agent Workflow

Describe the workflow of an AI Agent that monitors a clinic environment, analyzes operational conditions, detects bottlenecks, selects appropriate tools, executes actions, and generates operational summaries.

Prompt 7 - Project Challenges

Identify the major technical and operational challenges involved in developing an AI Agent for clinic operational management. Include simulation, synchronization, explainability, reliability, evaluation, and scalability challenges.

Prompt 8 - Success Metrics

Suggest quantitative success metrics for evaluating an AI Agent that manages clinic operations in a simulated healthcare environment.

Prompt 9 - Testing Scenarios

Generate realistic testing scenarios for evaluating an AI Agent responsible for clinic operational management, including department overload, medication shortages, room shortages, false-positive prevention, and full system integration testing.

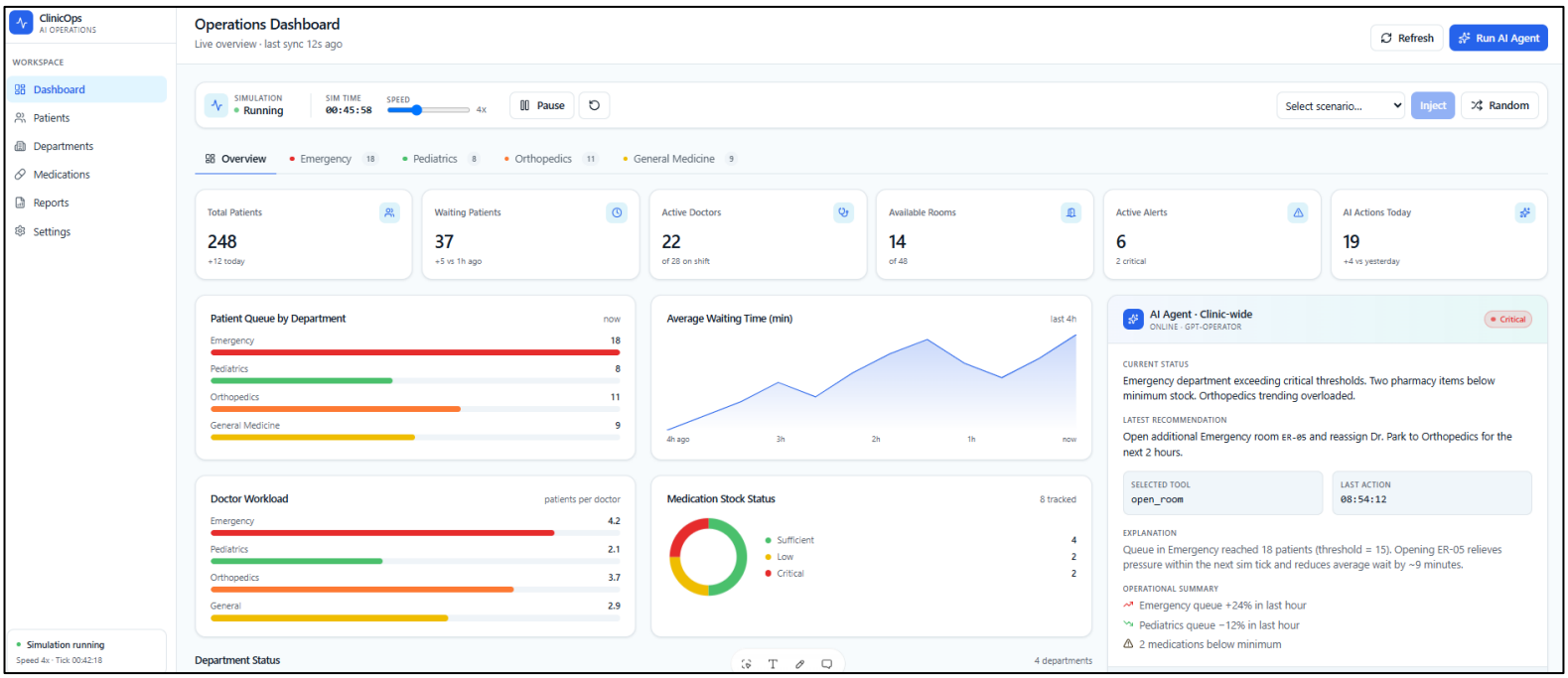
Prompt 10 - Research and Development Planning

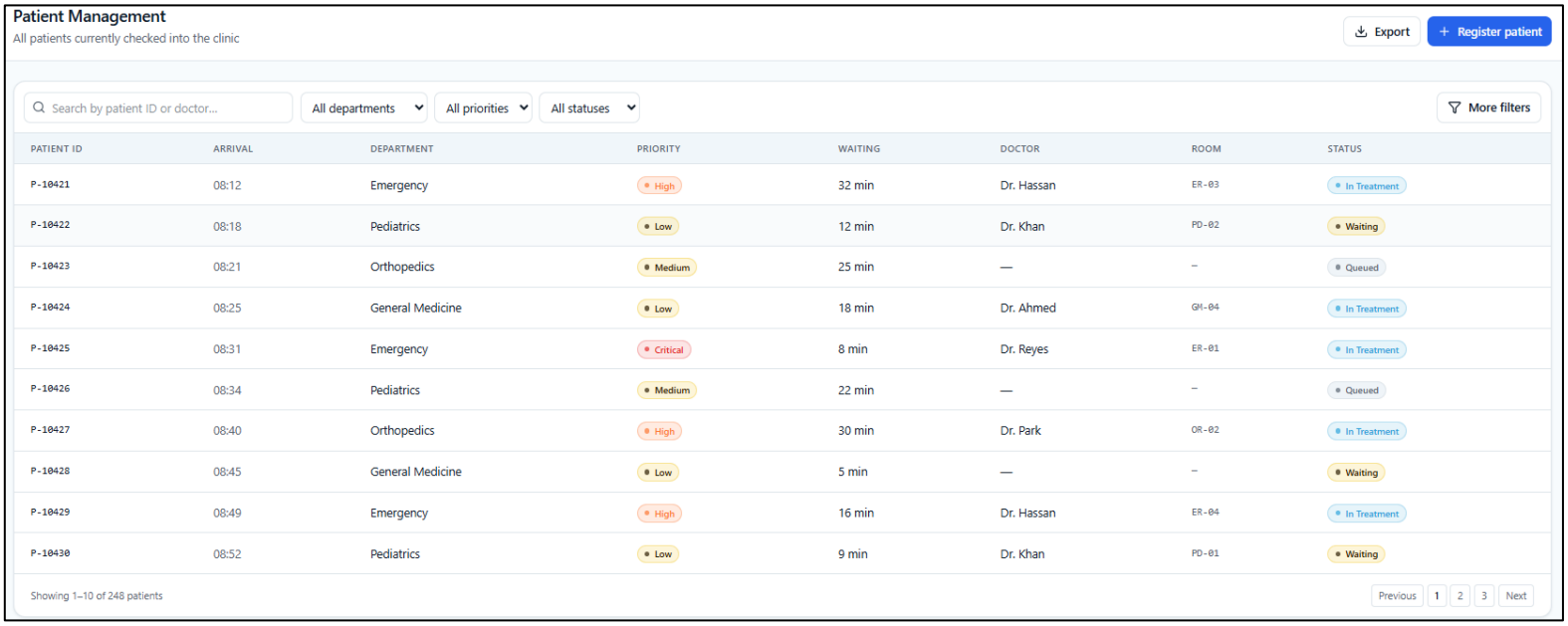
Create a structured research and development plan for a software engineering capstone project that develops an AI Agent for clinic operational management, including research activities, development stages, integration, testing, and evaluation.

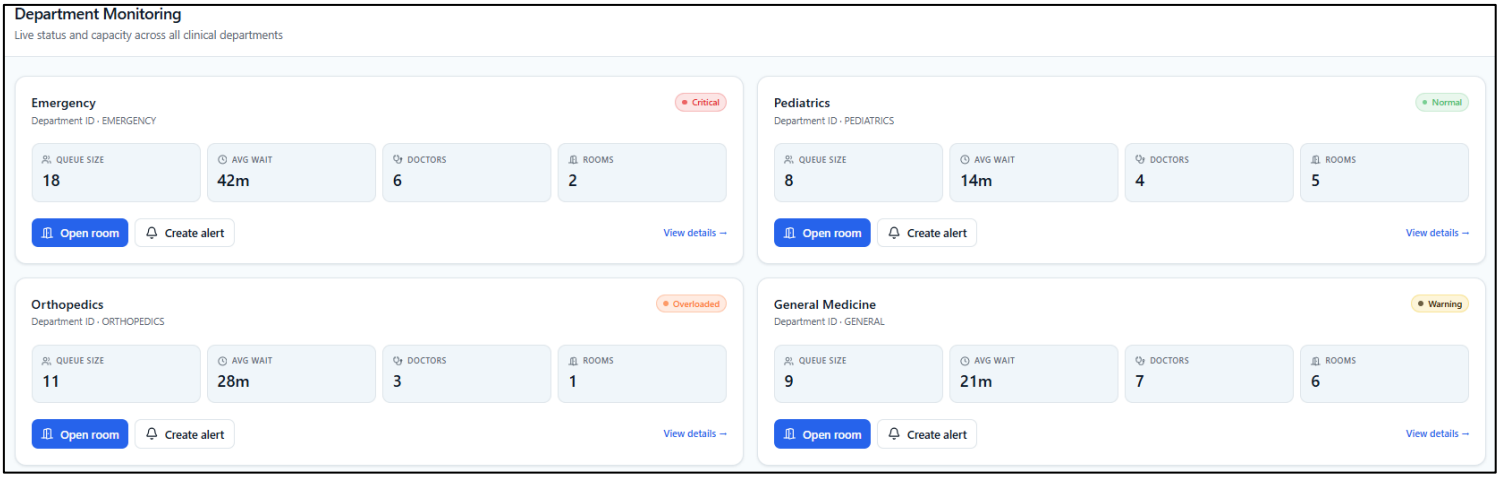
AI Usage Declaration

ChatGPT was used as a supporting tool for research, brainstorming, document organization, requirement formulation, architecture design, and technical writing. All project decisions, system design choices, requirements, diagrams, and final content were reviewed, validated, and adapted by the project team before inclusion in the project documentation.

11 - System UI/UX Specification







Medication Inventory

Stock levels across the clinic pharmacy

Request bulk restock

Total SKUs	Sufficient	Low	Critical
8	4	2	2

Search medications...

Filter by status

MEDICATION	CURRENT STOCK	MINIMUM THRESHOLD	STOCK LEVEL	STATUS	LAST RESTOCK	
Paracetamol 500mg	1,240	500		Sufficient	2026-06-24	Request restock
Amoxicillin 250mg	85	200		Low	2026-06-12	Request restock
Ibuprofen 400mg	640	300		Sufficient	2026-06-20	Request restock
Insulin Glargine	18	40		Critical	2026-06-05	Request restock
Salbutamol Inhaler	52	25		Sufficient	2026-06-18	Request restock
Adrenaline 1mg/mL	9	20		Critical	2026-06-02	Request restock
Ceftriaxone 1g	145	100		Sufficient	2026-06-22	Request restock
Morphine 10mg	28	30		Low	2026-06-15	Request restock

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